Embedding-Based vs. Probabilistic Topic Modeling: A Comparative Evaluation of BERTopic and LDA on the 20 Newsgroups Dataset

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Abstract

This study presents a comparative analysis of Latent Dirichlet Allocation (LDA) and BERTopic on the 20 Newsgroups dataset, with the aim of evaluating differences between probabilistic and embedding-based topic modelling approaches. LDA represents documents using a bag-of-words framework and models topics as latent word co-occurrence distributions, whereas BERTopic leverages transformer-based embeddings, clustering, and class-based TF-IDF , weighting to construct semantically coherent topic representations.

Both models were evaluated using intrinsic metrics, including topic coherence (c_v), topic diversity, topic distinctiveness, and intertopic Jaccard similarity, as well as clustering-based metrics such as Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI). The results indicate that LDA achieved slightly higher topic coherence, reflecting strong word co-occurrence modeling. However, BERTopic outperformed LDA in clustering alignment with ground-truth labels, achieving higher ARI and NMI scores, along with improved topic diversity and reduced inter-topic overlap.

Qualitative inspection further revealed that while LDA produced several interpretable domain-specific topics, it also generated more generic background themes. In contrast, BERTopic demonstrated stronger document-level semantic grouping and clearer topic separation. These findings suggest that embedding-based topic modelling provides advantages in capturing higher-level semantic structure, particularly for document clustering tasks, while LDA remains competitive in optimizing lexical coherence.

1 Introduction

Topic modelling provides a framework for identifying hidden semantic patterns by discovering groups of words that frequently occur together across documents. These methods are widely used in applications such as document organization, information retrieval, and exploratory data analysis.

Among topic modelling approaches, probabilistic generative models such as Latent Dirichlet Allocation (LDA) introduced by Blei et al. (2003). LDA assumes that documents are generated from mixtures of latent topics, where each topic is characterized by a distribution over words. Although LDA remains a commonly used baseline, recent advances in natural language processing have introduced embedding-based methods that incorporate contextual semantic representations.

One such approach is BERTopic, introduced by Grootendorst (2022), which combines transformer-based sentence embeddings with clustering techniques and class-based term weighting to generate interpretable topics. Unlike traditional probabilistic models, BERTopic leverages dense vector representations of documents to capture semantic similarity beyond simple word co-occurrence patterns.

This study presents a systematic comparison between LDA and BERTopic using the 20 Newsgroups dataset, a benchmark corpus containing documents from multiple discussion categories. The study evaluates both models using intrinsic metrics, including topic coherence, topic diversity, and topic distinctiveness, as well as extrinsic clustering metrics such as Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI) with respect to the ground-truth labels.

The objective of this study is to examine the relative strengths and limitations of probabilistic and embedding-based topic modelling approaches within a controlled experimental setting. By applying consistent preprocessing and evaluation procedures, this study aims to provide an empirical assessment of how model design influences topic interpretability and clustering quality.

2 Related Work

Topic modeling has evolved significantly since the introduction of Latent Dirichlet Allocation (LDA) by Blei et al. (2003). Traditional approaches like LDA, Non-Negative Matrix Factorization (NMF), and Probabilistic Latent Semantic Analysis (PLSA) rely on word co-occurrence statistics and bag-of-words representations. While effective for many applications, these methods often struggle with capturing semantic relationships and handling polysemy.

Recent advances in natural language processing, particularly transformer-based models like BERT Devlin et al. (2019), have enabled new approaches to topic modeling. BERTopic Grootendorst (2022) leverages sentence transformers to create dense document embeddings, followed by dimensionality reduction and clustering. This approach captures semantic relationships more effectively than traditional statistical methods.

Recent research, Liu (2024) conducted a comparative analysis of Latent Dirichlet Allocation (LDA) and BERTopic, focusing specifically on their efficacy in modeling topics from news content.This work aligns with the broader recognition of topic modeling's value in the digital transformation era, where machine learning and NLP are crucial for revealing patterns in large-scale textual data. Liu concludes by suggesting future research into hybrid modeling approaches and investigating the generalizability of these models across different domains and languages.

3 Dataset

In this studied used 20 Newsgroups Dataset. The Scikit-learn Developers (2013) represents one of the seminal collections in text classification and topic modeling research. The dataset captures a snapshot of early internet discussion culture through Usenet newsgroups distributed discussion systems that were precursors to modern internet forums. The collection comprises approximately 20,000 documents evenly distributed across 20 different thematic categories, making it an ideal testbed for evaluating topic modeling algorithms.

3.1 Preprocessing

The 20 Newsgroups dataset was loaded using the `fetch_20newsgroups` function from scikit-learn, with headers, footers, and quoted reply sections removed to reduce structural noise. This step ensures that the models focus on the substantive textual content rather than email metadata or formatting artifacts.To improve data quality and reduce sparsity, documents containing fewer than 50 tokens were excluded from further analysis.

Shared Basic Cleaning : A shared base cleaning procedure was applied prior to model-specific preprocessing to ensure consistency between LDA and BERTopic. The following steps were performed:

- Conversion of all text to lowercase.
- Removal of email addresses and URLs using regular expressions.
- Removal of alphanumeric tokens containing digits (e.g., "g9v", "a86"), which frequently occur in encoded or technical fragments.
- Removal of standalone numeric tokens.
- Removal of non-alphabetic characters, retaining only letters and whitespace.
- Normalization of excessive whitespace.

LDA-specific preprocessing included lemmatization, stopword removal, POS filtering, bigram detection, and vocabulary filtering.

BERTopic used cleaned documents without lemmatization to preserve contextual information.

4 Methods

4.1 Dataset

In this studied used 20 Newsgroups Dataset. Approximately 20,000 newsgroup documents partitioned across 20 different categories. Initial dataset inspection revealed 18,846 documents across 20 balanced categories.

4.2 Preprocessing

The 20 Newsgroups dataset, when properly preprocessed using the comprehensive pipeline described above, provides an excellent foundation for comparing topic modeling approaches.

4.3 Latent Dirichlet Allocation (LDA)

LDA is a Bayesian generative model where documents are mixtures of topics and topics are distributions over words. A 15-topic model was trained using Gibbs sampling.

4.4 BERTopic

BERTopic leverages the `all-MiniLM-L6-v2` transformer to generate 384-dimensional embeddings, followed by dimensionality reduction was performed using UMAP to preserve local semantic structure, followed by HDBSCAN clustering to identify groups of semantically similar documents. Finally, class-based TF-IDF was applied to extract representative keywords for each cluster by comparing term frequency within clusters against the entire corpus.

4.5 Evaluation Metrics

1. **Coherence (c_v) :** Coherence (c_v) measures the semantic consistency and interpretability of topics by evaluating how frequently the top words within a topic co-occur together in the original document corpus, using normalized pointwise mutual information (NPMI) and cosine similarity to quantify semantic relationships between topic words Röder et al. (2015) .

2. **Topic Diversity :** Topic Diversity measures the lexical variety across different topics by calculating the proportion of unique words among the top words of all topics, quantifying how much vocabulary overlap exists between topics.

3. **Topic Distinctiveness :** Topic Distinctiveness measures how well-separated different topics are from each other by calculating 1 minus the average pairwise Jaccard similarity between the top words of all topics, quantifying the conceptual separation between topics, and indicating whether topics represent truly distinct themes or overlapping concepts.

4. **Average Jaccard Similarity:** Average Jaccard Similarity directly measures topic overlap by calculating the mean pairwise Jaccard index between topic word sets, where the Jaccard index is the ratio of intersection to union of two sets, providing a direct quantification of lexical overlap between different topics across the entire model.

5. **Adjusted Rand Index (ARI) :** The Adjusted Rand Index (ARI) evaluates clustering quality by measuring pairwise agreement between predicted topic assignments and ground-truth labels, while correcting for chance agreement.

6. **Normalized Mutual Information (NMI) :** Normalized Mutual Information (NMI) quantifies the amount of shared information between the predicted and true label distributions using an information-theoretic framework.

5 Results

5.1 Result Summary

While LDA produced marginally higher topic coherence, BERTopic consistently outperformed LDA in terms of clustering alignment with the ground-truth categories, as reflected by higher ARI and NMI scores. BERTopic also generated more diverse and better-separated topics, with reduced overlap between topic word sets. These results suggest that embedding-based clustering methods may better capture broader semantic relationships across documents, whereas LDA focuses more narrowly on lexical co-occurrence patterns.

Metric	LDA	BERTopic
Coherence (c_v)	0.5888	0.5254
Topic Diversity	0.7800	0.8655
Topic Distinctiveness	0.9680	0.9808
Average Jaccard Similarity	0.0319	0.0191
Adjusted Rand Index	0.2508	0.3066
Normalized Mutual Information	0.4363	0.6548

Table 1: Metric Comparison

5.2 Topic word overlap heatmaps

To examine the degree of overlap between topics, Jaccard similarity scores were calculated using the top 15 words per topic. The similarity matrices are illustrated in Figures 1 and 2 through heatmap visualizations.

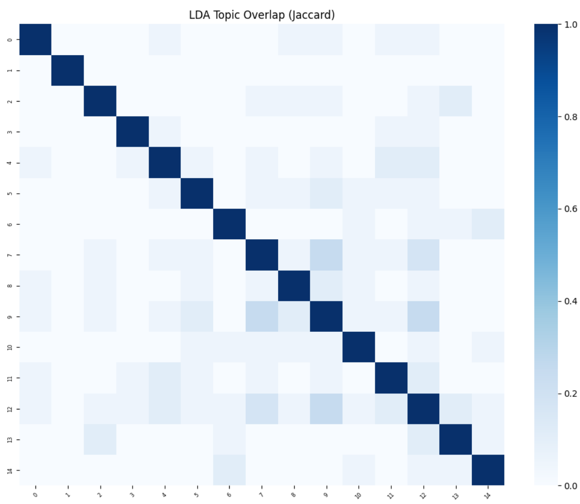

Figure 1: LDA topic word overlap heatmap

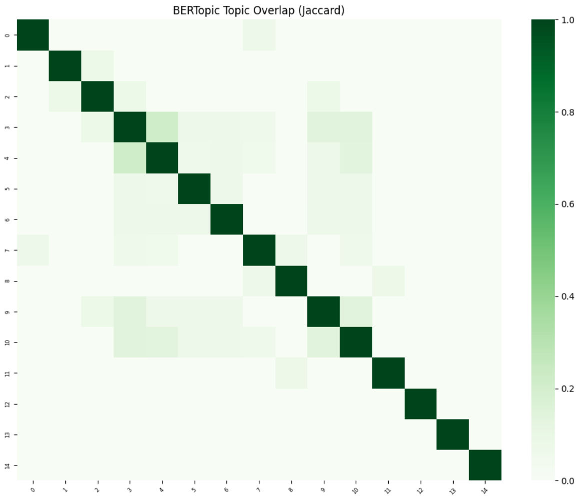

Figure 2: BERTopic topic word overlap heatmap

The Jaccard similarity heatmaps further illustrate structural differences between the models. BERTopic exhibits a strongly diagonal matrix with minimal off-diagonal intensity, indicating limited lexical overlap between topics and clear thematic separation. In contrast, the LDA heatmap shows more moderate off-diagonal similarity values, suggesting greater vocabulary sharing across topics. These visual patterns align with the quantitative results, where BERTopic achieved lower average Jaccard similarity and higher topic distinctiveness.

5.3 Manual Topic Inspection

Inspired by Chang et al. (2009).LDA produced several clearly interpretable domain-specific topics.In contrast, BERTopic produced more semantically cohesive clusters with reduced lexical overlap, reflecting improved document-level separation. This qualitative inspection aligns with the quantitative evaluation metrics, where BERTopic demonstrated higher topic diversity and stronger alignment with ground-truth categories. For example, one topic captured "nuclear"," energy" and "power"

6 Discussion

The results of this study demonstrate notable differences between probabilistic and embedding-based topic modelling approaches. While LDA achieved slightly higher topic coherence under the c_v metric, BERTopic consistently outperformed LDA on clustering-based evaluation measures, including Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI). Additionally, BERTopic exhibited higher topic diversity and lower inter-topic word overlap, indicating improved separation among discovered themes.

These findings suggest that the two models optimize different aspects of topic quality. LDA directly models word co-occurrence patterns within documents, which aligns closely with the c_v coherence metric. As a result, it is not surprising that LDA achieved marginally higher coherence scores. In contrast, BERTopic relies on transformer-based embeddings combined with dimensionality reduction and density-based clustering. This architecture emphasizes document-level semantic similarity rather than purely lexical co-occurrence patterns.

The superior ARI and NMI scores observed for BERTopic indicate stronger alignment between discovered topics and the ground-truth categories of the 20 Newsgroups dataset. This suggests that embedding-based clustering captures higher-level semantic structure more effectively than bag-of-words probabilistic modelling. These results are consistent with prior work by Grootendorst (2022), who demonstrated that embedding-based topic modelling can improve semantic grouping by leveraging contextual representations.

The coherence metric used in this study, c_v, is a widely adopted evaluation measure shown to correlate with human interpretability judgments Röder et al. (2015). However, as emphasized by Chang et al. (2009), coherence alone does not fully capture human topic interpretability. A model may achieve high coherence by grouping frequently co-occurring words, yet still fail to reflect meaningful document-level structure.

This distinction is reflected in the present results. Although LDA produced slightly more coherent word sets, qualitative inspection revealed that it also generated more generic or background topics characterized by high-frequency discussion terms. In contrast, BERTopic produced more semantically cohesive clusters with reduced lexical redundancy, which is consistent with its higher diversity and distinctiveness scores. These results suggest that embedding-based approaches may be particularly advantageous when the primary objective is document clustering or alignment with predefined categories, while probabilistic models remain competitive in generating tightly co-occurring word groups.

7 Conclusion

This study presented a comparative evaluation of Latent Dirichlet Allocation (LDA) and BERTopic on the 20 Newsgroups dataset using both intrinsic and clustering-based evaluation metrics. The objective was to examine how probabilistic topic modelling and embedding-based approaches differ in terms of topic coherence, diversity, distinctiveness, and alignment with ground-truth document categories.

The results indicate that LDA achieved slightly higher topic coherence (c_v), suggesting that it remains effective at modeling word co-occurrence patterns and producing internally consistent topic-word distributions. However, BERTopic consistently outperformed LDA on clustering-oriented metrics, including Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI), demonstrating stronger alignment with true document labels. Furthermore, BERTopic exhibited higher topic diversity and lower inter-topic word overlap, indicating improved separation between themes.

Qualitative inspection supported these findings. While LDA generated several clearly interpretable domain-specific topics, it also produced more generic background themes dominated by high-frequency terms. In contrast, BERTopic yielded semantically cohesive clusters with reduced lexical redundancy, reflecting its ability to capture contextual relationships at the document level through transformer-based embeddings.

Overall, the findings suggest that embedding-based topic modelling provides advantages when the primary objective is document clustering and semantic structure discovery. LDA, however, remains competitive in generating coherent word groupings grounded in statistical co-occurrence. Therefore, the choice between methods should be guided by the intended application: lexical interpretability versus semantic clustering performance.

Limitations

Despite the comprehensive evaluation conducted in this study, several limitations should be acknowledged.

First, the analysis was performed on a single dataset (20 Newsgroups). While this dataset is widely used for benchmarking topic models, its structure and clearly defined categories may favor clustering-based approaches such as BERTopic. Therefore, the findings may not fully generalize to other domains, particularly shorter texts, noisy social media data, or highly specialized corpora.

Second, model hyperparameters were selected based on reasonable defaults rather than extensive hyperparameter optimization. For LDA, the number of topics, passes, and Dirichlet priors influence topic quality. Similarly, BERTopic performance is sensitive to the choice of embedding model, UMAP dimensionality reduction parameters, and HDBSCAN clustering settings. Different configurations may yield different comparative results.

Third, the evaluation relied primarily on automated metrics such as c_v coherence, topic diversity, ARI, and NMI. Although coherence has been shown to correlate with human judgments, automated measures do not fully capture interpretability or usefulness in real-world applications. A human evaluation study would provide a more robust assessment of topic quality.

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